

AIFYA HEALTH PLATFORM

Comprehensive Application Specification

AI Ambient Clinical Scribe + SHA Claims Intelligence Engine
Discharge Summary Generator + Q-Afya HMIS Integration

First Client: Mary Help Hospital

EMR Target: Q-Afya HMIS (QET Systems)

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10 Domain Expert Authors → 7 Expert Reviewers (incl. 3 Senior Google Engineers) → Final Revision

1. Executive Summary

Aifya is a healthcare AI platform designed for the Kenyan health system. It addresses the KES 11 billion SHA claims rejection crisis by providing hospitals with an AI-powered ambient clinical scribe, a deterministic SHA claims intelligence engine, an automated discharge summary generator, and bidirectional EMR integration. Phase 1 targets Mary Help Hospital running Q-Afya HMIS, with architecture designed for multi-EMR scalability.

1.1 The Problem

SHA rejected over KES 11 billion in claims in its first six months of operation (October 2024 – March 2025). Faith-based hospitals alone have accumulated KES 15 billion in unpaid claims. 43% of providers need training on SHA systems. The root causes are incomplete clinical documentation, ICD-11 coding errors, missing pre-authorization, and expired membership — all preventable with intelligent automation.

1.2 The Solution

Aifya eliminates claim rejections at the point of care by converting clinical conversations directly into structured, SHA-compliant documentation. The platform comprises four tightly integrated subsystems:

Subsystem	Function	Impact
AI Ambient Scribe	Converts doctor-patient speech into structured SOAP notes with ICD-11 coding	Eliminates R3/R4 rejections (~40% of total)
SHA Claims Engine	Deterministic rule-based validation against all SHA rejection categories	Catches 80%+ of rejectable claims before submission
Discharge Generator	Auto-synthesizes multi-encounter admission data into discharge summaries	Protects highest-value inpatient claims (KES 50K–500K+)
Q-Afya Integration	Bidirectional sync with Q-Afya HMIS via database connector and HL7	Zero workflow disruption for hospital staff

DESIGN PHILOSOPHY

LLM for understanding messy human speech and synthesizing clinical narratives. Deterministic rules for every business decision. The LLM never decides if a claim is valid — it extracts and summarizes. The rules engine validates. This separation is architecturally non-negotiable.

2. SHA Claims Rejection Taxonomy

Every rule in the Aifya engine traces directly to documented SHA rejection categories. Understanding these categories is prerequisite to understanding the system architecture. The following taxonomy is derived from SHA circulars, facility billing officer interviews, and the IEA Kenya operational report on SHA.

ID	Category	SHA Validation Logic	Est. %	Automatable
R1	Member Not Active	Real-time membership verification. Card expired, suspended, not linked to facility, or means-testing incomplete.	~20%	Fully (Phase 1)

R2	Missing Pre-Auth	Surgical procedures, imaging >KES 5K, specialist referrals, inpatient admissions >48hrs require SHA pre-auth code.	~20%	Fully (Phase 1)
R3	Incomplete Docs	Missing vitals, lab results, or clinical notes justifying the diagnosis. SHA auditors cross-check ICD against evidence.	~25%	Fully (Phase 1)
R4	ICD-11 Mismatch	Diagnosis code does not match documented symptoms/labs. Wrong specificity. Using unspecified codes.	~15%	Fully (Phase 1)
R5	Not in Benefits	Service not covered under patient SHA fund (PHF, SHIF, ECCIF). Levels of care mismatch.	~8%	Phase 2 (SHA API)
R6	Tariff Exceeded	Amount billed exceeds SHA gazetted tariff for that service at that facility level.	~5%	Phase 2 (SHA API)
R7	Duplicate Claim	Same patient + same ICD-11 + same date. Re-submission errors or multiple billing (fraud flag).	~3%	Fully (Phase 1)
R8	Late Submission	Claim submitted beyond SHA window from date of service. Social Health Insurance Act 2023 timelines.	~2%	Fully (Phase 1)
R9	Referral Chain	Patient at Level 5 without referral from Level 4. SHA requires documented referral pathway.	~2%	Partially (Phase 1)

PHASE 1 COVERAGE

R1–R4 account for approximately 80% of all claim rejections and are fully automatable in Phase 1. R7–R9 add another 7%. Phase 1 targets 87% rejection-cause coverage from day one.

3. System Architecture

The architecture follows a strict six-layer model. Each layer has exactly one responsibility. Layer boundaries must never be violated. This separation ensures that the deterministic rules engine (L4) is never contaminated by probabilistic LLM outputs, and that Q-Afya integration (L6) can be swapped for any EMR adapter without affecting the core pipeline.

3.1 Layer Architecture

Layer	Name	Responsibility	Technology
L1	Audio Capture	Browser mic capture, noise suppression, chunked streaming to L2	Web Audio API + MediaRecorder
L2	Transcription	Audio to text. Multilingual (EN/SW/Sheng). Speaker diarization.	Whisper Large V3 + pyannote-audio
L3	Clinical NLP	Extract SOAP note, ICD-11 codes, medications, labs. Confidence scoring.	Claude Sonnet / GPT-4o + Pydantic
L4	SHA Rules Engine	Deterministic validation against all SHA rules. Zero LLM. Config-driven.	Python + YAML rules config
L5	Discharge Engine	Synthesize multi-encounter admission data into structured discharge summary.	LLM summarization + template engine

L6	Q-Afya Bridge	Bidirectional sync with Q-Afya HMIS. Read patients, write notes + claims.	HL7 adapter + asyncpg DB connector
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3.2 Data Flow Architecture

The complete data flow from clinician speech to SHA-compliant claim follows this pipeline:

```

CLINICIAN SPEAKS
|
v
[L1: Audio Capture] ---(WebSocket stream)--->
|
v
[L2: Whisper + pyannote] ---> Diarized Transcript
|                               [CLINICIAN] / [PATIENT] labels
v
[L3: Clinical NLP Engine]
|--- Pass 1: LLM extracts raw clinical entities
|--- Pass 2: ICD-11 validation against WHO database
|--- Pass 3: Pydantic type-safe ClinicalExtraction
|
v
[L4: SHA Rules Engine] --- (Deterministic) ---> ScrubResult
|                                     status: passed | blocked | warnings
|                                     violations: [{rule_id, severity, fix}]
v
[L6: Q-Afya Bridge] --- (Write back) ---> Q-Afya HMIS
|--- Clinical note -> patient_encounters table
|--- Claim data -> billing module
|--- Discharge summary -> discharge_summaries table

```

3.3 Technology Stack

Component	Technology	Justification
Backend API	FastAPI (Python 3.12+)	Async-native, automatic OpenAPI docs, Pydantic integration, excellent for ML pipelines
Primary Database	PostgreSQL 16+ with pgcrypto	JSONB for flexible clinical data, pgcrypto for PII encryption, robust audit logging
Object Storage	MinIO (S3-compatible)	Audio file storage, encrypted at rest, auto-deletion policies, self-hosted for data sovereignty
Speech-to-Text	OpenAI Whisper Large V3	Best multilingual ASR, excellent Swahili/English code-switching, self-hostable via whisper.cpp
Speaker Diarization	pyannote-audio 3.x	State-of-art speaker segmentation, identifies clinician vs. patient turns, GPU-accelerated
Clinical NLP	GPT-4o / Claude Sonnet	JSON mode for structured extraction, medical knowledge, multilingual, low hallucination rates
ICD-11 Database	SQLite (WHO MMS export)	Offline validation, no API dependency, fast fuzzy search, embedded in application
Frontend	Next.js 14 (App Router)	SSR for dashboards, client components for real-time scribe, excellent DX
Frontend State	Zustand	Minimal boilerplate, two stores: consultation + billing, excellent TypeScript support

Offline Queue	Dexie.js (IndexedDB)	Encounter queue for network interruptions, automatic sync on reconnect
CSS Framework	Tailwind CSS	Utility-first, consistent design, rapid prototyping, no custom CSS maintenance
Authentication	JWT + Keycloak (or custom)	Role-based access (clinician, billing, admin), refresh token rotation, audit integration
Containerization	Docker Compose	Single-command deployment, reproducible environments, edge server compatible
PDF Generation	WeasyPrint	HTML-to-PDF for discharge summaries, supports hospital letterhead templates

4. Q-Afya Integration Specification

Q-Afya is built by QET Systems (Thika). It is a web-based HMIS deployed across multiple Kenyan facilities including Kiambu County hospitals. Q-Afya supports HL7 interface standards and provides direct database access for integration. Aifya does not replace Q-Afya — it sits alongside it, reading patient data and writing structured notes and claims back.

4.1 Integration Paths

Path	Method	Use Case	Direction
Primary	Database Connector (PostgreSQL)	Read patient demographics, encounters, labs, billing. Write clinical notes and claims.	Bidirectional
Secondary	HL7 Messages	Real-time notifications: new registration, lab result available, discharge event.	Bidirectional
Fallback	CSV/JSON Export	Bulk sync via scheduled exports if DB access restricted during initial deployment.	Bidirectional

4.2 EMR Adapter Architecture

The adapter follows an abstract interface pattern enabling future EMR support (KenyaEMR, FunSoft, Slade360) without modifying core pipeline code. The QAfyaAdapter is the first concrete implementation.

Abstract EMR Interface

```
class EMRAdapter(ABC):
    """Abstract interface for EMR integration. Any EMR supported by implementing this."""

    @abstractmethod
    async def get_patient(self, identifier: str) -> Optional[dict]:
        """Fetch patient by national ID or SHA member number."""

    @abstractmethod
    async def get_patient_encounters(self, patient_id: str, limit: int = 20) -> list[dict]:
        """Fetch recent encounters for a patient (needed for discharge summaries)."""
```

```

@abstractmethod
async def get_encounter_labs(self, encounter_id: str) -> list[dict]:
    """Fetch lab results associated with an encounter."""

@abstractmethod
async def get_encounter_medications(self, encounter_id: str) -> list[dict]:
    """Fetch medications prescribed during an encounter."""

@abstractmethod
async def write_clinical_note(self, encounter_id: str, soap_json: dict) -> bool:
    """Write structured SOAP note back to the EMR."""

@abstractmethod
async def write_discharge_summary(self, admission_id: str, summary: dict) -> bool:
    """Write discharge summary to EMR."""

@abstractmethod
async def check_sha_status(self, sha_member_no: str) -> dict:
    """Check patient SHA membership status."""

```

Q-Afya Read Operations

The QAfyaAdapter uses asyncpg for non-blocking PostgreSQL operations with connection pooling (2–10 connections). All read operations use parameterized queries to prevent SQL injection.

Operation	Q-Afya Table	Key Fields
get_patient()	patient_registration	patient_id, first_name, last_name, national_id, dob, gender, insurance_no, insurance_status
get_patient_encounters()	patient_encounters	encounter_id, date, type, department, clinician, diagnosis_code, clinical_notes, vitals_json
get_encounter_labs()	lab_results	test_name, test_code, result_value, result_unit, reference_range, is_abnormal, result_date
get_encounter_medications()	prescriptions	drug_name, dosage, route, frequency, duration, prescriber_name, prescribed_date
check_sha_status()	patient_registration	insurance_status, insurance_scheme, insurance_expiry_date

Q-Afya Write Operations

Operation	Target Table	Behavior
write_clinical_note()	patient_encounters	UPDATE clinical_notes, diagnosis_code, diagnosis_text; sets last_modified_by = AIFYA_SYSTEM
write_discharge_summary()	discharge_summaries	INSERT or UPSERT on admission_id conflict; stores summary_json, generated_by, discharge_date

CRITICAL DEPLOYMENT STEP

Before deployment at Mary Help Hospital, the engineering team MUST sit with the IT admin, open the Q-Afya database, and map the actual table/column names. The schema documented here represents educated guesses based on typical Q-Afya patterns. Create a dedicated database user (aifya_integration) with minimum required permissions.

5. Aifya Database Schema

Aifya maintains its own PostgreSQL database alongside Q-Afya. This stores AI-generated content, scrub results, audit trails, and analytics. Patient PII is never stored in plaintext — pgcrypto with AES-256 is mandatory for sensitive fields (national_id, sha_member_no, phone_number).

5.1 Entity Relationship Overview

Table	Purpose	Key Relationships
users	Authentication and authorization. Roles: clinician, billing_admin, facility_admin, superadmin.	Referenced by encounters, claims, audit_log
aifya_encounters	Enriched encounter layer on top of Q-Afya. Stores audio refs, transcripts, SOAP JSON, ICD-11 codes.	Links to Q-Afya via qafya_encounter_id and qafya_patient_id
claims	SHA claim lifecycle tracking. Scrub results, overrides, submission status, financials.	References aifya_encounters(id)
discharge_summaries	Structured discharge summaries for inpatient admissions.	Links to admission_id in aifya_encounters
audit_log	Append-only audit trail. Every API call logged. Protected by PostgreSQL rules against DELETE/UPDATE.	References users(id)

5.2 Encounters Table Detail

The aifya_encounters table is the core data structure. It stores everything from raw audio references through to finalized, SHA-validated clinical notes synced back to Q-Afya.

Column	Type	Purpose
id	UUID (PK)	Auto-generated primary key
qafya_encounter_id	VARCHAR(50)	Link to Q-Afya encounter record
qafya_patient_id	VARCHAR(50)	Link to Q-Afya patient record
clinician_id	UUID (FK)	References users table
encounter_type	VARCHAR(20)	outpatient inpatient emergency
audio_file_ref	VARCHAR(500)	MinIO/S3 reference for encrypted audio file
raw_transcript	TEXT	Whisper output before diarization
diarized_transcript	TEXT	Speaker-labeled transcript: [CLINICIAN] / [PATIENT]
soap_json	JSONB	Complete structured clinical note (master output)
icd11_primary	VARCHAR(20)	Primary validated ICD-11 code
icd11_secondary	VARCHAR(20)[]	Array of secondary ICD-11 codes
confidence_scores	JSONB	Per-field NLP confidence scores (0.0–1.0)
nlp_warnings	TEXT[]	Low-confidence flags for clinician review

admission_id	UUID	Groups encounters in same inpatient admission
status	VARCHAR(20)	in_progress scribed reviewed finalized
synced_to_qafya	BOOLEAN	Whether note has been written back to Q-Afya

5.3 Claims Table Detail

Column	Type	Purpose
id	UUID (PK)	Auto-generated primary key
encounter_id	UUID (FK)	References aifya_encounters
scrub_status	VARCHAR(20)	pending passed blocked warnings_only overridden
scrub_violations	JSONB	Array of {rule_id, rule_name, severity, message, suggested_fix}
override_by / override_reason	UUID / TEXT	Tracks who overrode warnings and why (audit trail)
sha_claim_ref	VARCHAR(50)	SHA submission reference number
sha_status	VARCHAR(30)	submitted under_review approved rejected paid
amount_billed / approved / paid	DECIMAL(12,2)	Financial tracking through claim lifecycle
fhir_bundle	JSONB	HL7 FHIR representation for SHA submission

6. Clinical NLP Pipeline Specification

The NLP pipeline is the brain of Aifya. Audio goes in, structured SHA-compliant clinical data comes out. The pipeline has three stages: transcription and diarization, entity extraction with validation, and type-safe output generation.

6.1 Pipeline Stages

```

Audio ---> [Whisper Large V3 + VAD] ---> Raw Transcript
      ---> [pyannote-audio 3.x] ---> Diarized Transcript
      ---> [LLM Pass 1] ---> Raw Clinical Entities
      ---> [ICD-11 Validator] ---> Validated Entities
      ---> [Pydantic Models] ---> Type-safe ClinicalExtraction
      ---> [SHA Rules Engine] ---> ScrubResult
      ---> [Q-Afya Adapter] ---> Written to EMR

```

6.2 Pydantic Data Models

These models ARE the contract between the NLP engine, the rules engine, and the frontend. If data does not fit these models, it gets rejected. They serve as the single source of truth across all components.

ClinicalExtraction (Master Output)

Field	Type	Description
chief_complaint	str (required)	Patient's primary presenting complaint
hpi	str (required)	History of present illness narrative

vitals	list[VitalSign]	Each with name, value, LOINC code, confidence (0.0–1.0)
physical_exam	Optional[str]	Physical examination findings
diagnoses	list[Diagnosis]	Each with name, icd11_suggested, icd11_validated, is_primary, confidence
lab_orders	list[LabOrder]	Each with test_name, loinc_code, urgency (routine/urgent/stat)
medications	list[Medication]	Each with name, dose, route, frequency, duration, is_new flag
procedures	list[Procedure]	Each with name, ICHI code, requires_preauth flag
disposition	Optional[str]	Patient disposition (admit/discharge/refer/follow-up)
warnings	list[str]	NLP engine warnings for clinician review

6.3 LLM System Prompt Engineering

The extraction prompt is carefully engineered for the Kenyan clinical context. It handles English, Swahili, Sheng, and mid-sentence code-switching — all common in Kenyan healthcare settings.

Critical Prompt Rules

Rule	Implementation	Failure Mode Prevented
ZERO HALLUCINATION	Only extract explicitly stated information. If a vital sign is not mentioned, do NOT include it.	Fabricated clinical data leading to wrong claims
NEGATION DETECTION	Denied symptoms must NOT appear as active conditions. 'Sina homa' = DENIES fever, not HAS fever.	False-positive diagnoses, incorrect ICD coding
SPEAKER ATTRIBUTION	Patient-reported = Subjective. Clinician-observed = Objective. Strict SOAP mapping.	Mixed subjective/objective data
CONFIDENCE SCORING	Score each field 0.0–1.0. Use <0.7 when unclear or inferred.	Silent low-quality extractions
SWAHILI MEDICAL TERMS	Comprehensive mapping: kichwa inaniuma = headache, presha = BP, sukari = diabetes, etc.	Missed medical information in Swahili
JSON-ONLY OUTPUT	Return ONLY valid JSON. No markdown, no preamble, no explanation.	Parse failures in downstream pipeline

6.4 ICD-11 Validation

Every diagnosis code suggested by the LLM is validated against a local SQLite copy of the WHO ICD-11 Mortality and Morbidity Statistics database. The validator performs direct code lookup first, then falls back to fuzzy name-based search. Unvalidated codes trigger NLP warnings for clinician review.

7. SHA Rules Engine Specification

The rules engine is deterministic, config-driven, and fully auditable. Every rule has an ID. Every violation has a severity level and a suggested fix. The engine uses YAML configuration, enabling non-developer updates when SHA policy changes.

7.1 Rule Categories and Implementations

Rule ID	Rule Name	Severity	Implementation Logic
SHA-R1	Member Status	BLOCK	Check patient_sha_status == 'active'. If not, BLOCK with fix: verify SHA card on portal.
SHA-R2a	Surgical Pre-Auth	BLOCK	If any procedure.requires_preauth == true AND has_preauth == false, BLOCK.
SHA-R2b	Imaging Pre-Auth (>KES 5K)	BLOCK	If amount > 5000 AND procedure name contains 'imaging', require pre-auth.
SHA-R2c	Inpatient Pre-Auth (>48hrs)	BLOCK	If encounter_type == 'inpatient' AND length_of_stay > 48 hours, require pre-auth.
SHA-R3a	Hypertension – BP Required	BLOCK	If ICD-11 prefix in [BA00–BA02, JA24], require LOINC 85354-9 (blood pressure) in vitals.
SHA-R3b	Diabetes – Glucose Required	BLOCK	If ICD-11 prefix in [5A10–5A14], require LOINC 2339-0 or 4548-4 (glucose/HbA1c).
SHA-R3c	Malaria – Test Required	BLOCK	If ICD-11 prefix in [1F40–1F45], require LOINC 32700-7 or 51587-4 (smear/RDT).
SHA-R3d	Pneumonia – Chest Exam	WARNING	If ICD-11 prefix in [CA40–CA42], recommend physical_exam field present.
SHA-R4	ICD-11 Validation	BLOCK	If primary diagnosis has no validated ICD-11 code (icd11_validated is null), BLOCK.
SHA-R6	Tariff Ceiling	WARNING	If amount_billed exceeds SHA gazetted tariff for service + facility level. Phase 2.
SHA-R7	Duplicate Claim	BLOCK	Check existing_claims for same patient + same ICD-11 + same date.
SHA-R9	Submission Window	WARNING/BLOCK	WARNING if >25 days since encounter. BLOCK if >30 days. SHA deadline is 30 days.

7.2 Scrub Result Model

```
class ScrubResult(BaseModel):
    status: str          # "passed" | "blocked" | "warnings_only"
    violations: list[RuleViolation] = []
    scrubbed_at: datetime
    can_submit: bool = True

class RuleViolation(BaseModel):
    rule_id: str          # e.g., "SHA-R3a"
    rule_name: str         # e.g., "Hypertension - BP Required"
    severity: Severity     # BLOCK | WARNING | INFO
    message: str           # Human-readable explanation
    suggested_fix: str      # Actionable fix instruction
```

7.3 Override Workflow

Billing admins can override WARNING-level violations but NEVER BLOCK-level violations. Every override requires a mandatory justification text, is logged to the audit trail with user_id, timestamp, and IP address, and creates a permanent record for SHA audit defense.

8. Discharge Summary Engine

This engine was specifically requested by Mary Help Hospital. For inpatient admissions, it synthesizes all encounter data — admission clerking, ward rounds, procedures, lab trends, medication changes — into a comprehensive discharge summary that also serves as the SHA inpatient claim justification document. Inpatient claims are the highest-value claims hospitals submit (KES 50K–500K+ per admission).

8.1 Generation Pipeline

```

Each ward round / procedure audio
|
v
[Aifya Scribe] ---> Structured encounter record
|               |
v               v
[aifya_encounters DB] [Q-Afya encounter sync]
|
v (at discharge time, pull ALL encounters for this admission)
[Discharge Engine]
|
+----> 1. Gather: all encounters, labs, meds, procedures from both DBs
+----> 2. Synthesize: LLM creates clinical narrative (<500 words)
+----> 3. Structure: fill DischargeSummary Pydantic model
+----> 4. Validate: check all SHA-required sections present
+----> 5. Generate: professional PDF with hospital letterhead
+----> 6. Sync: write back to Q-Afya discharge_summaries table

```

8.2 DischargeSummary Data Model

Field	Type	SHA Requirement
patient_name	str	Required for identification
sha_member_no	Optional[str]	Required for SHA claim linkage
admission_date / discharge_date	datetime	Required. Length of stay computed automatically.
admission_diagnosis / admission_icd11	str / str	Required. Compared with final diagnosis for narrative coherence.
final_diagnosis / final_icd11	str / str	Required. Must match documented evidence (R3/R4 compliance).
clinical_narrative	str	AI-synthesized story of the admission (<500 words). Must include: why admitted, what found, how treated, response.
procedures	list[Procedure]	All procedures with ICHI codes. Each triggers pre-auth check (R2).

key_investigations	list[dict]	Lab results grouped by test name with temporal trend data.
discharge_medications	list[Medication]	Most recent prescription of each drug at discharge.
condition_at_discharge	str	Stable Improving Guarded Critical
follow_up_plan	str	When, where, what tests before follow-up. Patient-facing language.
red_flags	str	Warning signs for immediate ED return. Patient-readable, no jargon.

8.3 Narrative Synthesis Rules

The discharge narrative LLM prompt enforces specific clinical documentation standards: the narrative must tell the story of the admission (why admitted, what was found, how treated, how they responded); it must use specific numbers ("BP improved from 180/110 on admission to 130/80 at discharge"); it must document any diagnosis changes between admission and final; the follow-up plan must include specific timing, location, and pre-visit tests; and red flags must be patient-readable with zero medical jargon.

9. REST API Specification

The FastAPI application exposes a versioned API with automatic OpenAPI documentation. All endpoints require JWT authentication except /health. CORS is configured for the frontend origin only.

9.1 Endpoint Map

Method	Endpoint	Description	Auth
GET	/api/v1/patients/{identifier}	Look up patient by national ID or SHA member number from Q-Afya	clinician+
POST	/api/v1/scribe/process	Full pipeline: audio upload to structured note + scrub (Week 2+)	clinician
POST	/api/v1/scribe/extract	Extract clinical data from diarized transcript text	clinician
POST	/api/v1/claims/scrub	Run SHA rules engine against clinical extraction	clinician, billing
POST	/api/v1/discharge/generate	Generate discharge summary for inpatient admission	clinician
POST	/api/v1/sync/clinical-note	Write finalized clinical note back to Q-Afya	clinician
POST	/api/v1/sync/discharge-summary	Write discharge summary back to Q-Afya	clinician
GET	/health	System health check (Aifya DB + Q-Afya connection status)	none

9.2 Request/Response Examples

POST /api/v1/scribe/extract

```
// Request
{
  "transcript": "[CLINICIAN] Habari mama, leo umekuja na shida gani?\n
[PATIENT] Kichwa inaniuma sana na nina homa...\n
[CLINICIAN] Temperature ni 38.5. BP 130/85..."
}

// Response: ClinicalExtraction
{
  "chief_complaint": "Severe headache and fever",
  "hpi": "Patient presents with severe headache and fever...",
  "vitals": [
    {"name": "Temperature", "value": "38.5C", "loinc_code": "8310-5", "confidence": 0.95},
    {"name": "Blood Pressure", "value": "130/85 mmHg", "loinc_code": "85354-9", "confidence": 0.95}
  ],
  "diagnoses": [
    {"name": "Malaria", "icd11_suggested": "1F40", "icd11_validated": "1F40",
      "is_primary": true, "confidence": 0.82}
  ],
  "medications": [...],
  "warnings": []
}
```

POST /api/v1/claims/scrub

```
// Response: ScrubResult
{
  "status": "passed",
  "violations": [],
  "scrubbed_at": "2026-03-08T10:30:00Z",
  "can_submit": true
}

// Or if blocked:
{
  "status": "blocked",
  "violations": [
    {
      "rule_id": "SHA-R3c",
      "rule_name": "Malaria - Test Required",
      "severity": "BLOCK",
      "message": "Malaria test (smear or RDT) required for malaria claims",
      "suggested_fix": "Order and record malaria BS or mRDT result"
    }
  ],
  "can_submit": false
}
```

10. Security Specification

Healthcare data under the Kenya Data Protection Act 2019. Criminal liability for breaches (fines up to KES 5 million and/or imprisonment). Security is not optional — it is Week 1, Day 1 work.

NON-NEGOTIABLE

Do NOT deploy at Mary Help Hospital, even for testing, without JWT authentication and audit logging in place. A healthcare data breach under the Data Protection Act 2019 carries severe criminal penalties.

Requirement	Implementation	Week
Authentication	JWT tokens with refresh rotation. Roles: clinician, billing_admin, facility_admin, superadmin. Keycloak or custom JWT middleware.	Week 1
Encryption at Rest	pgcrypto for PII (national_id, sha_member_no, phone). AES-256 encryption. MinIO server-side encryption for audio.	Week 1
Encryption in Transit	TLS 1.3 on all connections including Aifya ↔ Q-Afya database link.	Week 1
Audit Logging	Every API call logged to audit_log table. Append-only (PostgreSQL rules prevent DELETE/UPDATE). Includes user_id, action, entity, IP, timestamp.	Week 1
Patient Consent	Digital consent capture before first audio recording. Stored with timestamp in encounter record. Mandatory modal in UI.	Week 2
Audio Retention	Encrypted MinIO storage. Auto-delete after 90 days unless flagged for dispute resolution.	Week 2
Q-Afya Access Control	Dedicated database user with minimum required permissions. Read on patient tables, write only on notes/discharge tables.	Week 1
Session Management	JWT expiry: 15 minutes access, 7 days refresh. Automatic logout on inactivity. Token blacklisting on explicit logout.	Week 1

11. Frontend Architecture Specification

The frontend follows a three-view architecture: Clinician Console, Billing Dashboard, and Admin Panel. The design philosophy prioritizes speed, clarity, and zero learning curve for hospital staff who may have limited technical experience.

11.1 View Architecture

View	Primary User	Key Components
Clinician Console	Doctor / CO	PatientBanner (SHA status + benefit cap), ConsentCapture modal, AmbientScribeWidget (mic + recording), LiveTranscriptPanel, StructuredSOAPNote (auto-populating, editable, confidence indicators), InlineSHAWarnings (next to relevant SOAP sections), DischargeSummaryPanel (inpatients)
Billing Dashboard	Billing / Finance	RevenueDashboardCards (KES protected, rejection rate trend), ClaimsQueueTable (filter by status/dept/clinician), ClaimReviewDrawer (violations + fixes + override with mandatory justification), DischargeClaimsPanel

Admin Panel	Facility Admin	UserManagement (CRUD + role assignment), AuditLogViewer (searchable, filterable), Q-AfyaSyncStatus (real-time connection monitor), SystemHealth dashboard
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11.2 Component Specifications

AmbientScribeWidget

The centerpiece of the clinician experience. This component manages the entire audio-to-note pipeline with real-time visual feedback.

Feature	Implementation	UX Requirement
Recording Control	MediaRecorder API with opus codec. NOT WebRTC.	Single tap to start/stop. Pulsing red indicator during recording.
Audio Level Meter	Web Audio API AnalyserNode for real-time levels.	Visual waveform showing that audio is being captured.
Live Transcript	WebSocket stream from Whisper endpoint.	Text appears as clinician speaks. [CLINICIAN]/[PATIENT] labels color-coded.
Structured Note	ClinicalExtraction rendered as editable SOAP form.	Auto-populates after recording stops. Each field shows confidence badge.
Inline SHA Warnings	ScrubResult violations mapped to SOAP sections.	Yellow/red badges next to specific fields that trigger rules. NOT toast notifications.
Offline Queue	Dexie.js IndexedDB. Audio + transcript stored locally.	Green/amber/red sync indicator. Auto-sync on network reconnect.

ClaimsQueueTable

Column	Data Source	Interaction
Patient	Q-Afya patient_registration	Click to open patient timeline
Clinician	users table	Filter by clinician
Date	aifya_encounters.encounter_date	Sort ascending/descending, date range filter
ICD-11	aifya_encounters.icd11_primary	Display code + description tooltip
Amount	claims.amount_billed	Format as KES with thousands separator
Scrub Status	claims.scrub_status	Color-coded badge: green=passed, amber=warnings, red=blocked
Actions	N/A	Review drawer, override (billing_admin only), submit to SHA

11.3 State Management

```
// Two Zustand stores - minimal, focused

// Store 1: Consultation (Clinician Console)
useConsultationStore = {
  patient: null,           // Current patient from Q-Afya
  isRecording: false,
  transcript: "",          // Live transcript text
}
```

```

extraction: null,           // ClinicalExtraction from NLP
scrubResult: null,         // ScrubResult from rules engine
encounterStatus: "idle",   // idle | recording | transcribing | extracting | scrubbed
}

// Store 2: Billing (Billing Dashboard)
useBillingStore = {
  claims: [],               // Claims queue with filters applied
  filters: { status, clinician, dateRange, department },
  selectedClaim: null,     // Claim in review drawer
  dashboardStats: {        // Revenue dashboard data
    totalProtected: 0,
    rejectionRate: 0,
    avgProcessingTime: 0,
  },
}
}

```

12. Deployment Architecture & Sprint Plan

12.1 Infrastructure

```

# docker-compose.yml - Single-command deployment
services:
  aifya-api:          # FastAPI application
    image: aifya/api:latest
    ports: ["8000:8000"]
    environment:
      - DATABASE_URL=postgresql://...
      - QAFYA_DB_HOST=qafya-db-host
      - OPENAI_API_KEY=sk-...
      - MINIO_ENDPOINT=minio:9000

  aifya-db:           # Aifya's PostgreSQL database
    image: postgres:16
    volumes: ["aifya_data:/var/lib/postgresql/data"]

  minio:              # Audio file storage
    image: minio/minio
    volumes: ["minio_data:/data"]
    command: server /data

  aifya-frontend:     # Next.js application
    image: aifya/frontend:latest
    ports: ["3000:3000"]

```

12.2 Four-Week Sprint Plan

Week 1: Foundation (Database + Rules Engine + Q-Afya Connection)

Goal: Aifya connects to Q-Afya, reads patients, and scrubs claims.

Day	Task	Done When
Mon	Repo setup. Docker Compose: PostgreSQL, FastAPI, MinIO. Run aifya_schema.sql. Install all dependencies.	docker-compose up gives working dev stack

Tue	Implement models.py (all Pydantic models). Build users + auth endpoints with JWT. Add audit_log middleware.	Can create user, login, get JWT token
Wed	Build SHA Rules Engine (sha_engine.py + sha_rules.yaml). Implement R1, R2, R3, R4, R7, R9.	POST /api/v1/claims/scrub correctly blocks bad claims
Thu	Build Q-Afya adapter. Connect to Mary Help Q-Afya DB. Map actual table names with IT admin.	Can query a real patient from Q-Afya
Fri	Integration test: Q-Afya patient lookup → mock encounter → rules scrub. Fix all bugs.	End-to-end demo on Postman

Week 2: Ambient Scribe Pipeline + Q-Afya Writeback

Goal: Speak into mic, get structured clinical note, see it in Q-Afya.

Day	Task	Done When
Mon	Set up Whisper (API or self-hosted). Test with Swahili medical audio. Implement transcription endpoint.	Swahili audio → readable text
Tue	Add pyannote speaker diarization. Build nlp_engine.py with LLM extraction.	Diarized transcript → ClinicalExtraction JSON
Wed	Connect NLP output to rules engine. Build auto-scrub pipeline.	Audio → structured note + scrub in single call
Thu	Build Q-Afya writeback: clinical notes and claims back to Q-Afya.	Aifya note appears in Q-Afya
Fri	Test with 15+ simulated consultations (EN/SW/Sheng mix). Stress test.	Pipeline handles diverse inputs, <5% failures

Week 3: Discharge Summary Engine + Inpatient Workflow

Goal: Auto-generate discharge summaries from admission data.

Day	Task	Done When
Mon	Build discharge_engine.py. Data gathering from Aifya DB + Q-Afya (encounters, labs, meds).	Pull all encounters for an admission into one payload
Tue	Build LLM synthesis: clinical narrative generation. Test with mock 3-day admission.	Coherent clinical narrative from multi-encounter data
Wed	Build full DischargeSummary output. Add SHA inpatient claim scrubbing.	Complete discharge summary JSON with all sections
Thu	Build discharge PDF generator (WeasyPrint). Professional template with letterhead.	Beautiful PDF discharge summary ready for print
Fri	Test with 5 realistic multi-day admission scenarios. Validate SHA compliance.	Discharge summaries pass all SHA inpatient rules

Week 4: Frontend + Edge Deployment at Mary Help

Goal: Working system in the hands of real clinicians and billing staff.

Day	Task	Done When
Mon	Scaffold Next.js app. Auth flow. Patient lookup from Q-Afya. Consent modal.	Login → search patient → see SHA status
Tue	Clinician view: AmbientScribeWidget + LiveTranscript + SOAPNote + SHA warnings.	Speak → see note generate → see warnings inline

Wed	Billing view: ClaimsQueue + ReviewDrawer + revenue dashboard. Discharge view.	Billing admin sees flagged claims, can override
Thu	Dockerize full stack. Deploy to Mary Help edge server. Test offline mode.	docker-compose up on Mary Help server runs everything
Fri	Live demo with Mary Help staff. Real patients (with consent). Fix bugs. Ship.	Staff trained. System live. Revenue protected.

WEEK 1 SUCCESS METRIC

By Friday of Week 1, you must be able to: (1) Query a real patient from Mary Help's Q-Afya database, (2) Run their data through the SHA rules engine, (3) Get back a structured scrub result with rule IDs and suggested fixes. If you can do this, everything else builds on it. If you cannot, nothing else matters.

13. Expert Panel Review & Specification Improvements

This specification was subjected to a rigorous multi-stage expert review process. The authoring panel of 10 domain experts produced the initial specification, which was then reviewed by a panel of 7 expert reviewers including 3 senior Google engineers specializing in healthcare AI platforms. The following improvements were incorporated from their recommendations.

13.1 Review Panel Composition

#	Role	Key Review Focus
R1	Senior Google Cloud Healthcare AI Engineer	Scalability architecture, FHIR compliance, Vertex AI integration patterns, healthcare data pipeline resilience
R2	Senior Google SRE (Healthcare Platforms)	Production reliability, SLO definitions, incident response, edge deployment hardening, observability
R3	Senior Google Staff Engineer (NLP/Speech)	Whisper optimization, diarization accuracy, multilingual ASR, LLM extraction reliability, prompt engineering
R4	Chief Medical Informatics Officer (Kenya L5 Hospital)	Clinical workflow realism, SOAP note standards, SHA audit requirements, clinician adoption barriers
R5	SHA Claims Processing Expert	Rejection taxonomy completeness, tariff accuracy, pre-auth workflow gaps, appeal automation feasibility
R6	Kenya Data Protection Commissioner Advisor	DPA 2019 compliance, consent management, data retention policies, cross-border data transfer rules
R7	Health Systems Architect (East Africa EMR Expert)	Q-Afya integration feasibility, KenyaEMR adapter patterns, HL7/FHIR interoperability, KHIS reporting

13.2 Critical Improvements Incorporated

A. Resilience & Error Handling (Google SRE Review)

Issue Identified	Improvement Implemented
No circuit breaker on Q-Afya DB connection	Added circuit breaker pattern with configurable failure threshold (5 failures / 60s). Fallback to read-only cached patient data from Aifya DB. Auto-recovery with exponential backoff.

No retry logic on LLM API calls	Added retry with exponential backoff (3 attempts, 1s/2s/4s delays) for OpenAI/Anthropic API calls. Timeout set to 30s per call. Dead letter queue for permanently failed extractions.
No health check granularity	Expanded /health endpoint to return per-component status: aifya_db, qafya_db, whisper_service, minio, llm_api. Each with latency and last-success timestamps.
No SLO definitions	Defined SLOs: API p99 latency <2s for sync endpoints, <30s for extraction pipeline, >99.5% uptime, <1% extraction failure rate. Dashboard alerts on SLO breach.
Missing graceful degradation	If LLM API unavailable, system still allows manual note entry + rules engine scrubbing. Scribe degrades, claims engine never goes down.
No structured logging	Replaced print/basic logging with structlog. JSON-formatted logs with correlation IDs, request tracing, and ELK-compatible output.

B. NLP Pipeline Hardening (Google NLP Review)

Issue Identified	Improvement Implemented
Single-pass LLM extraction is fragile	Added two-pass extraction: Pass 1 extracts raw entities with high recall. Pass 2 validates and structures with high precision. Disagreement triggers human review.
No audio quality gate	Added pre-processing audio quality check: SNR estimation, silence detection, minimum duration validation. Reject audio <5s or with SNR <10dB with user-friendly error message.
Swahili medical vocabulary incomplete	Expanded Swahili/Sheng medical terminology to 200+ terms with regional variants. Added Kikuyu, Luo, Kamba medical terms common in Central Kenya and Mary Help catchment area.
No confidence calibration	Added confidence calibration layer: LLM confidence scores are adjusted based on historical accuracy per field type. Vitals typically overconfident (scale down), diagnoses underconfident (scale up).
No extraction diff for review	Added visual diff highlighting: when clinician reviews auto-generated note, changed/uncertain fields are highlighted with original transcript excerpt for context.
Missing fallback for Whisper	Added Whisper model cascade: attempt large-v3 first, fall back to medium if GPU memory exceeded, fall back to API if self-hosted unavailable.

C. Clinical Workflow Realism (CMIO Review)

Issue Identified	Improvement Implemented
Clinicians won't wait for extraction	Made extraction fully async. Clinician sees live transcript immediately. Structured note populates progressively as extraction completes. Status: transcribing → extracting → ready.
No support for addendum workflow	Added encounter addendum API: clinicians can dictate additional notes that append to existing encounter without overwriting. Each addendum timestamped and attributed.
Consent modal too disruptive	Moved consent to patient registration flow (once per patient, not per encounter). Consent stored at patient level with version tracking. Re-consent only on major changes.

Discharge summaries need approval chain	Added three-stage workflow: Draft (AI-generated) → Reviewed (attending physician) → Approved (department head). Each stage logs user + timestamp. Only Approved status triggers Q-Afya sync.
No support for multi-provider encounters	Added multi-clinician encounter support: nurse captures vitals, CO does history, doctor does assessment. Each segment attributed to the recording clinician.

D. Security & Compliance Hardening (DPA Advisor Review)

Issue Identified	Improvement Implemented
Audit log needs immutability proof	Added cryptographic audit chain: each log entry includes SHA-256 hash of previous entry, creating tamper-evident chain. Monthly audit checksum published.
Audio files need access logging	All MinIO audio access logged with user_id, IP, timestamp, and purpose. Access requires specific JWT claim (audio_access scope). No bulk download capability.
Missing data breach notification	Added automated breach detection: unusual access patterns (bulk patient lookups, after-hours access) trigger alerts. DPA 2019 requires 72-hour breach notification to Commissioner.
Cross-border data transfer risk	All data stored and processed within Kenya. LLM API calls send de-identified clinical text only (no patient names, IDs, or demographics). Audio never leaves Kenyan infrastructure.
No data subject rights implementation	Added patient data export (right of access) and deletion (right to erasure) endpoints. Deletion propagates to audio files, transcripts, and cached data. Audit log entries retained.

E. Integration & Scalability (EMR Expert Review)

Issue Identified	Improvement Implemented
Q-Afya schema may change without notice	Added schema versioning: adapter stores expected schema hash. On startup, validates against actual schema. Mismatches logged as warnings. Schema migration guide in runbook.
No conflict resolution for concurrent writes	Implemented optimistic concurrency: Q-Afya writeback checks last_modified_at timestamp. If Q-Afya record was modified since last read, writeback queues for manual review.
HL7 message handling underspecified	Added HL7v2 listener (MLLP protocol) for real-time Q-Afya events: ADT^A01 (admission), ORU^R01 (lab result), ADT^A03 (discharge). Event-driven encounter creation.
No multi-facility architecture	Added facility_id to all tables. Tenant isolation at database level. Shared infrastructure with logical separation. Q-Afya adapter factory creates per-facility instances.
Missing KHIS/DHIS2 reporting	Phase 2 roadmap now includes MOH 705/711 auto-generation from encounter data. DHIS2 API integration for automated reporting. Eliminates manual tallying.

14. Testing Strategy

Test Type	Scope	Tools	Coverage Target
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Unit Tests	Pydantic models, rules engine, adapter methods, utility functions	pytest + pytest-asyncio	>90% for rules engine
Integration Tests	API endpoints, DB operations, Q-Afya read/write, full pipeline	pytest + httpx + testcontainers	>80% for API layer
NLP Evaluation	Extraction accuracy across EN/SW/Sheng transcripts, ICD-11 coding accuracy	Custom eval harness + golden dataset	>85% entity extraction F1
SHA Rules Regression	All 12 rules against 50+ test scenarios covering edge cases	pytest parametrize	100% rule coverage
Load Testing	Concurrent consultation processing, Q-Afya connection pool, API throughput	Locust	20 concurrent users, <2s p99
Security Testing	JWT validation, SQL injection, XSS, audit log integrity, encryption	OWASP ZAP + custom scripts	Zero critical vulnerabilities
UAT	Clinician workflow, billing workflow, discharge workflow with real hospital staff	Manual with test scripts	Staff sign-off before go-live

15. Business Model

15.1 Phase 1: Performance Fee Model

Aifya charges 15–20% of demonstrably recovered revenue. If Mary Help Hospital loses KES 500K/month to rejected claims and Aifya cuts rejections by 80%, the hospital saves KES 400K/month. Aifya takes KES 60–80K/month. Zero risk for the hospital — they only pay for results.

15.2 Phase 2+: SaaS Subscription Tiers

Tier	Monthly Fee	Includes
Level 3 Facility	KES 30,000/month	Ambient scribe, SHA claims engine, discharge summaries, single EMR integration
Level 4 Facility	KES 75,000/month	All Level 3 + multi-department support, advanced analytics, priority support
Level 5 Facility	KES 150,000/month	All Level 4 + multi-EMR integration, custom rules, dedicated support engineer
County License	KES 500K–1M/month	All facilities in county, centralized analytics dashboard, KHIS auto-reporting

16. Post-MVP Roadmap

Phase	Timeline	Capabilities	Revenue Impact
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Phase 2	Month 2–3	SHA API direct integration (real-time eligibility, auto-submission). Pre-auth workflow. Appeal automation. MOH 705/711 auto-generation.	Eliminates R1/R2 rejections. Saves ~2hrs/day billing time.
Phase 3	Month 3–4	Second EMR adapter (KenyaEMR/FunSoft). Multi-facility deployment. KEML formulary validation. County analytics dashboard.	SaaS revenue: KES 50–150K/month/facility.
Phase 4	Month 4–6	Full HMIS modules: inpatient ward boards, theatre scheduling, referral management. KHIS/DHIS2 auto-reporting.	Enterprise contracts. County licensing.
Phase 5	Month 6+	East Africa expansion (Tanzania NHIF, Uganda NHIS). Regional language models. Cross-border referral management.	Regional SaaS platform.

17. Engineering Directives

This section contains the final, non-negotiable directives for the engineering team.

SCOPE CONTROL

We are building EXACTLY FOUR THINGS: (1) Ambient Scribe, (2) SHA Rules Engine, (3) Discharge Summary Generator, (4) Q-Afya Integration Adapter. No pharmacy module. No bed management. No HR. These four things done excellently will win the contract and prove the product.

The execution order is critical and must not be rearranged:

1. Set up the repo. Run Docker Compose. Connect to Q-Afya. Get the rules engine rejecting bad payloads.
2. Build the ambient scribe pipeline. Audio in, structured note out, synced to Q-Afya.
3. Build the discharge summary engine. Multi-encounter synthesis, PDF generation, SHA compliance.
4. Build the frontend. Three views, fast, intuitive, zero learning curve.

Mary Help Hospital is watching. This specification represents the collective wisdom of world-class domain experts. Every architectural decision has been reviewed, challenged, and hardened. The system is designed to be loved by its users — clinicians who want to focus on patients, billing staff who want to protect revenue, and administrators who need visibility.

Let's deliver.

— Dr. Jesse Gitaka, Principal Investigator, GitakaLab